Using Zero and Negative Exponents

The zero exponent, rewriting a negative exponent as a fraction, and the laws when the result is zero or negative

Directions

- Any non-zero base to the power zero is 1. Dividing a power by itself proves it.
- A negative exponent means *reciprocal*, never a negative answer: $x^{-n} = \frac{1}{x^n}$ for $x \neq 0$.
- Final answers use positive exponents only, and numerical answers are exact fractions — not decimals.

1. Evaluate: 7^0.

Answer: ________________

2. Evaluate: $4^0 + 9^0$.

Answer: ________________

3. Evaluate as an exact fraction: 2^{-3}.

Answer: ________________

4. Evaluate as an exact fraction: 5^{-2}.

Answer: ________________

5. Simplify: $x^3 \cdot x^{-3}$.
What does the product law give you here, and what is that worth?

Answer: ________________

6. Simplify, given that n is not zero: $(2n)^0$.

Answer: ________________

7. Rewrite with a positive exponent: y^{-6}.

Answer: ________________

8. Simplify, and write the answer with a positive exponent: $\dfrac{x^2}{x^7}$.

Answer: ________________

9. Evaluate as an exact fraction: 10^{-4}.
This is the size of one ten-thousandth, the kind of factor that shows up in scientific notation.

Answer: ________________

10. Simplify, and write the answer with a positive exponent: $(3m^{-2})(4m^5)$.

Answer: ________________

11. Write <, >, or = between the two quantities. Give the value of each in your work.

$$6^0 \quad \underline{\hspace{2cm}} \quad 6^{-1}$$

12. **Challenge.** Dara writes $3^{-2} = -9$.

(a) Give the correct value as an exact fraction. Answer: _________

(b) Explain what the negative exponent actually does, and why the answer cannot be negative.